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Roll No

FT-705 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Chemical Safety
Time : Three Hours

Maximum Marks : 70

- Note:** i) Answer any five questions.
ii) All questions carry equal marks.

1. How toxicants enter biological organisms?
2. Describe the different organs of Respiratory system and explain the effect of chemical exposure on Respiratory system for following gases:
 - i) Ammonia (NH_3)
 - ii) Chlorine (Cl_2)
3. What do you understand by Hypersensitivity? Explain the different types of hypersensitivity with their causes and symptoms.
4. Enlist the details description on input data and output data for Risk Analysis Methodology available for System Safety Assessment.
5. Give the detail description on human lungs structure?
6. Define Hazard Risk and Risk Reduction? Write the process of managing the risk in process plant with suitable block diagram.

OR

Define hazardous material? Give the detail classification and sub classification based on the united nation classification system.

7. Demonstrate the model for mass flow rate for the following condition of source model.
 - i) Flow of liquid through a hole
 - ii) Flow of liquid through a hole in a tank
 - iii) Flow of vapor through hole
 - iv) Flow of flashing liquids
8. Answer any four of the following:
 - a) Explain the inhalation effects of NH_3 on human body.
 - b) What are limitations of Pasquill-Gifford model?
 - c) Explain the stability of any chemical constituent.
 - d) What is fundamental difference between flammable and combustible stock.
 - e) Give the short description on FTA.
 - f) Explain Event Tree Analysis (ETA).
